

3 (a) State the principle of moments.

(for a system in equilibrium,) sum of clockwise moments (about a point) equals sum of anticlockwise moments (about the same point)	B1
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..... [1]

(b) A rigid uniform beam rests on a pivot at its centre, as shown in Fig. 3.1.

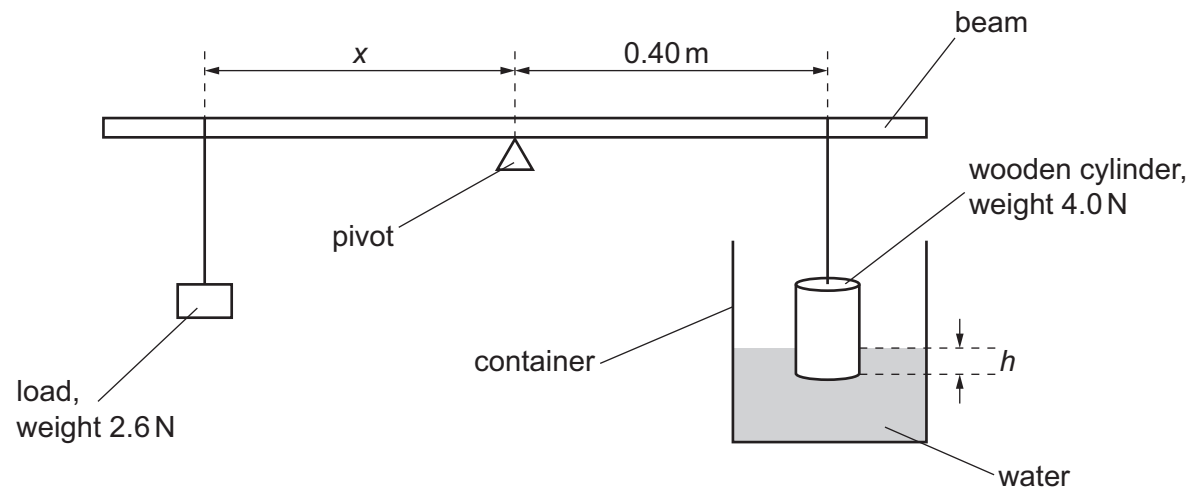


Fig. 3.1 (not to scale)

A load of weight 2.6 N is suspended from the beam at distance x from the pivot.

A wooden cylinder of weight 4.0 N is suspended from the beam at a distance of 0.40 m from the pivot on the opposite side of the pivot to the load. The cylinder rests in a container of water. The lower part of the cylinder is immersed in the water to depth h .

Initially, h is equal to 0.10 m and x is equal to 0.40 m . The system is in equilibrium.

(i) Use the principle of moments to show that the upthrust U exerted by the water on the cylinder is 1.4 N .

2.6×0.40 or $F \times 0.40$ (any one moment)	C1
$U = 4.0 - F$ $= 4.0 - (2.6 \times 0.40 / 0.40) = 1.4\text{ N}$	A1
or	
2.6×0.4 or $(4.0 - U) \times 0.4$	(C1)
$2.6 \times 0.4 = (4.0 - U) \times 0.4$ hence $U = (4.0 - 2.6) = 1.4\text{ N}$	(A1)

[2]

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(ii) The density of the water is $1.0 \times 10^3 \text{ kg m}^{-3}$.

Calculate the area A of the circular cross-section of the cylinder.

$U = \rho g V$ and $A = V/h$ or $p = h\rho g$ and $A = U/p$ or $A = U/h\rho g$	C1
$A = 1.4 / (0.10 \times 1.0 \times 10^3 \times 9.81)$	C1
$= 1.4 \times 10^{-3} \text{ m}^2$	A1

$A = \dots\dots\dots \text{ m}^2$ [3]

(c) More water is gradually added to the container in (b), so that depth h in Fig. 3.1 gradually increases. The length x is continuously adjusted so that the system remains in equilibrium.

On Fig. 3.2, sketch the variation of x with h . Use the space below for any working.

line starting at (0.10, 0.40)	B1
straight line with negative gradient	B1
line ending at (0.29, 0)	B1

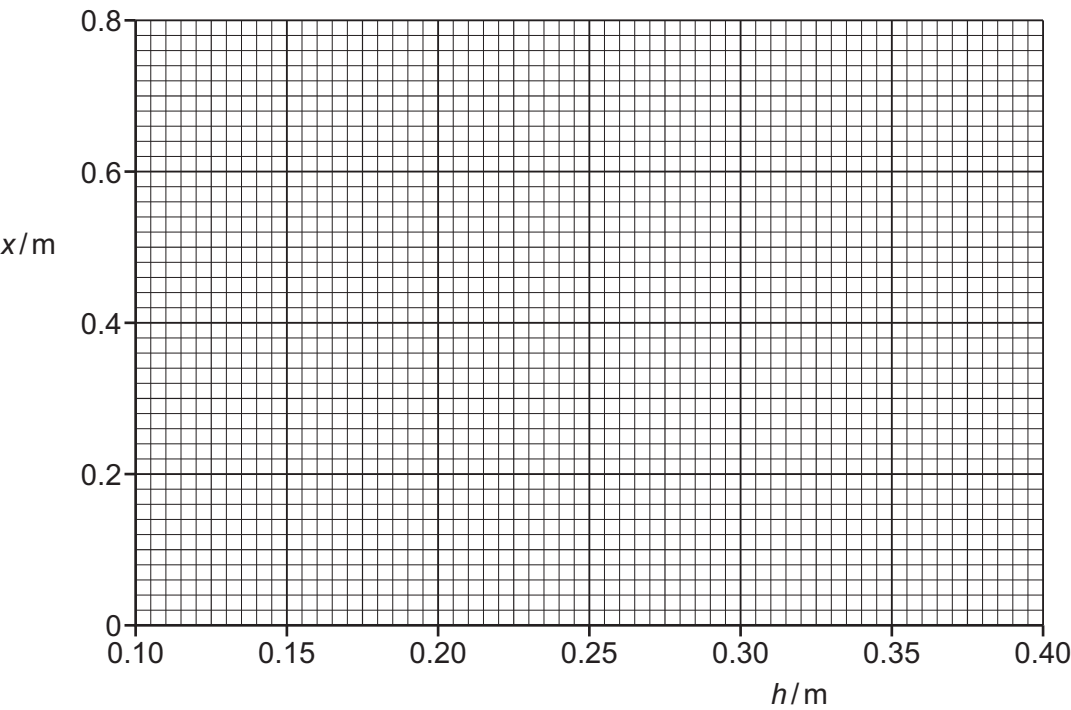


Fig. 3.2

[3]

[Total: 9]

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- 4 A pinball machine uses a spring to launch a small metal ball of mass $4.5 \times 10^{-2} \text{ kg}$ up a ramp. The spring is compressed by $8.0 \times 10^{-2} \text{ m}$ and held in equilibrium, as shown in Fig. 4.1.

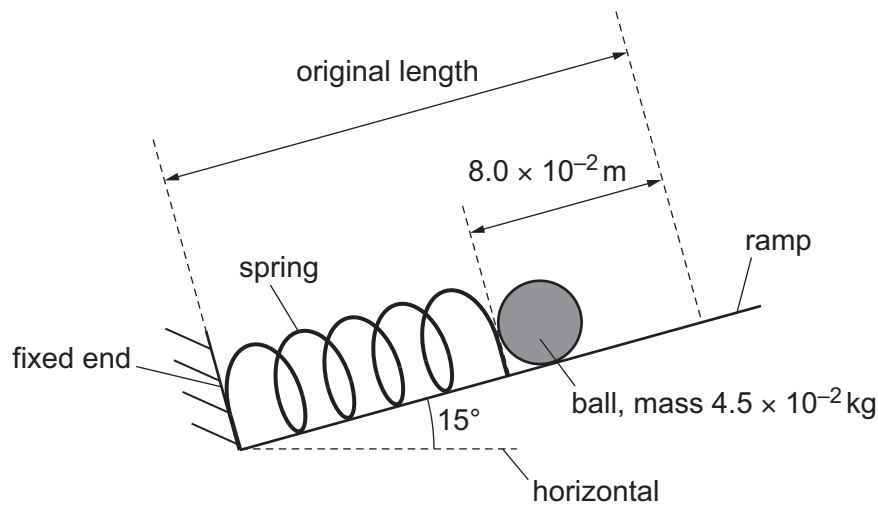


Fig. 4.1 (not to scale)

The ramp is at an angle of 15° to the horizontal.

- (a) The spring obeys Hooke's law and has a spring constant of 29 N m^{-1} .

Calculate the elastic potential energy in the compressed spring.

$E = \frac{1}{2}kx^2$ or $E = \frac{1}{2}Fx$ and $F = kx$	C1
$E = \frac{1}{2} \times 29 \times (8.0 \times 10^{-2})^2$ or $E = \frac{1}{2} \times 2.32 \times 8.0 \times 10^{-2}$ $E = 9.3 \times 10^{-2} \text{ J}$	A1

elastic potential energy = J [2]

- (b) The spring is released and expands quickly back to its original length.

- (i) Calculate the increase in gravitational potential energy of the ball when the spring returns to its original length.

$(\Delta)E_{(P)} = mg(\Delta)h$	C1
$= 4.5 \times 10^{-2} \times 9.81 \times 8.0 \times 10^{-2} \sin 15^\circ$	C1
$= 9.1 \times 10^{-3} \text{ J}$	A1

increase in gravitational potential energy = J [3]

2024 Paper 2 Questions by Topic

- (ii) The ball leaves the spring when the spring reaches its original length. Assume that all the elastic potential energy of the spring is transferred to the ball.

Calculate the speed of the ball as it leaves the spring.

$E_{(K)} = \frac{1}{2}mv^2$	C1
$(9.3 \times 10^{-2} - 9.1 \times 10^{-3}) = \frac{1}{2} \times 4.5 \times 10^{-2} \times v^2$	C1
$v = (2 \times 8.4 \times 10^{-2} / 4.5 \times 10^{-2})^{0.5}$ $= 1.9 \text{ m s}^{-1}$	A1

speed = m s^{-1} [3]

- (c) The ball comes to rest on a horizontal trapdoor of negligible mass at a distance d from its pivot.
A force F acts vertically downwards at a distance of 2.0 cm from the pivot, as shown in Fig. 4.2.

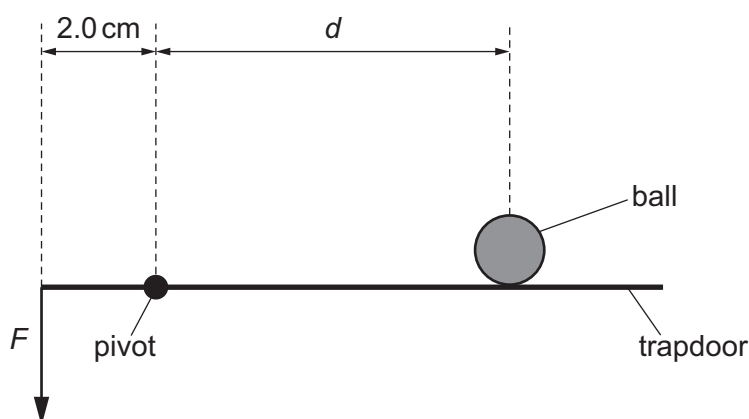


Fig. 4.2 (not to scale)

- (i) The trapdoor is in equilibrium when F is 1.7 N.

Calculate d .

$1.7 \times 2.0 (\times 10^{-2})$ or $4.5 \times 10^{-2} \times 9.81 \times d$	C1
$1.7 \times 2.0 \times 10^{-2} = 4.5 \times 10^{-2} \times 9.81 \times d$ $d = 7.7 \times 10^{-2} \text{ m}$	A1

$d =$ m [2]

- (ii) Force F is decreased from 1.7 N.

State the direction of the resultant moment about the pivot on the trapdoor.

clockwise	B1 [1]
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[Total: 11]

2024 Paper 2 Questions by Topic

- 3 (a)** State what is meant by the work done by a force.

product of force and displacement in direction of force	B1
.....	[1]

- (b)** A block of mass m is raised vertically at constant speed. The vertical height gained by the block is Δh , as shown in Fig. 3.1.

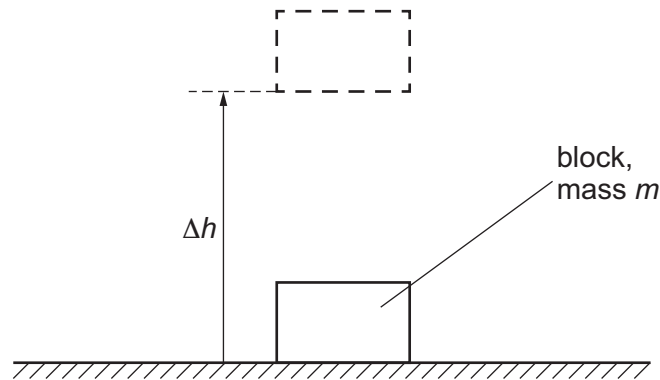


Fig. 3.1

Derive an expression, in terms of m and Δh , for the change in gravitational potential energy ΔE_p of the block. State the meaning of any other symbols you use.

weight/force = mg , and g identified as acceleration of free fall	B1
ΔE_p identified as work (done by lifting force/weight), and so = $mg \times \Delta h$	B1

[2]

- (c)** An electric motor has an input power of 900 W. The motor takes 1.0 minute to lift a load of weight 240 N at constant speed through a vertical height of 150 m. Resistive forces are negligible.

- (i)** Show that the work done by the motor on the load in 1.0 minute is 36 kJ.

$240 \times 150 = 36 \text{ kJ}$	A1
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[1]

2024 Paper 2 Questions by Topic

- (ii) Determine the useful output power of the motor.

$P = W/t$	C1
$= 36\,000 / 60$	A1
$= 600\text{ W}$	

power = W [2]

- (iii) Use your answer in (c)(ii) to determine the efficiency of the motor.

efficiency = <u>useful</u> output power / total input power	C1
$= 600 / 900$	A1
$= 0.67$	

efficiency = [2]

- (iv) Some of the power wasted in the motor is dissipated by the resistance of its coil. This dissipated power is 280 W.

The coil of the motor is made from wire of total length 23 m. The wire has a cross-sectional area of $2.6 \times 10^{-8} \text{ m}^2$ and is made from metal of resistivity $1.7 \times 10^{-8} \Omega \text{ m}$.

Calculate the current in the coil.

$P = I^2 R$ and $R = \rho L / A$	C1
$280 = I^2 \times (1.7 \times 10^{-8} \times 23) / (2.6 \times 10^{-8})$	C1
$I = 4.3\text{ A}$	A1

current = A [3]

[Total: 11]